Supplementary data 1 — Spider Web-Honeycomb Results

Supporting information for the article:

“Stochastic Spider webs: Bio-inspired Metamaterials Design and Structural Optimization”

Bayesian Optimization Results — 12 Cases

Mohamed Elleithy and Pınar Acar

E-mail: melleithy@vt.edu · pacar@vt.edu

Abstract

This document presents the complete Bayesian optimization results for twelve structural cases comparing spider web and honeycomb metamaterial plates under simply supported (SS) and clamped (CC) boundary conditions subjected to uniform and non-uniform transverse pressure. Three objectives are considered: maximum von Mises stress, maximum elastic strain energy density (ESEDEN), and total ESEDEN. For each case the optimal design parameters, performance metrics, Finite Elements field contours, and optimization convergence histories with Gaussian-process posterior uncertainty bands are presented.

Keywords: Metamaterials; Bayesian Optimization; Spider web; Honeycomb; Structural design; Structural optimization; Elastic energy density; von Mises stress; Uncertainty quantification

Smart Materials and Structures, 2026

GLOBAL PERFORMANCE SCORECARD

Table 1 summarizes the primary objective value, winner, and performance margin for all twelve cases. SW = spider web; HC = honeycomb.

#	Case	BC	Load	Objective	HC	SW	Winner	Margin
1	SS · Uniform · von Mises	Simply supported	Uniform pressure	Minimize maximum von Mises stress	749.3 MPa	542.6 MPa	SW	28%
2	SS · Uniform · Total ESEDEN	Simply supported	Uniform pressure	Minimize total ESEDEN	8677 MPa	5224 MPa	SW	40%
3	SS · Uniform · Max ESEDEN	Simply supported	Uniform pressure	Minimize maximum ESEDEN	0.578 MPa	0.374 MPa	SW	35%
4	SS · Non-Uniform · von Mises	Simply supported	Non-uniform pressure	Minimize maximum von Mises stress	350.6 MPa	364.1 MPa	HC	4%
5	SS · Non-Uniform · Total ESEDEN	Simply supported	Non-uniform pressure	Minimize total ESEDEN	1325 MPa	1034 MPa	SW	22%
6	SS · Non-Uniform · Max ESEDEN	Simply supported	Non-uniform pressure	Minimize maximum ESEDEN	0.1183 MPa	0.1868 MPa	HC	37%
7	CC · Uniform · von Mises	Clamped	Uniform pressure	Minimize maximum von Mises stress	639.7 MPa	641.2 MPa	TIE	0.2%
8	CC · Uniform · Total ESEDEN	Clamped	Uniform pressure	Minimize total ESEDEN	2090 MPa	570.8 MPa	SW	73%
9	CC · Uniform · Max ESEDEN	Clamped	Uniform pressure	Minimize maximum ESEDEN	0.3858 MPa	0.6112 MPa	HC	37%
10	CC · Non-Uniform · von Mises	Clamped	Non-uniform pressure	Minimize maximum von Mises stress	272.8 MPa	325.8 MPa	HC	16%
11	CC · Non-Uniform · Total ESEDEN	Clamped	Non-uniform pressure	Minimize total ESEDEN	269.9 MPa	115.3 MPa	SW	57%
12	CC · Non-Uniform · Max ESEDEN	Clamped	Non-uniform pressure	Minimize maximum ESEDEN	0.0765 MPa	0.1241 MPa	HC	38%

Table 1. *Summary scorecard for all 12 optimization cases.*

1. SS · Uniform · von Mises

BC: Simply supported · Load: Uniform pressure · Objective: Minimize maximum von Mises stress

Spider Web wins +28%

1.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.4378**
β	**0.13615**
t	**8 mm**
h	**30.09 mm**

Area fraction: **57.56%** · Shell thickness: **8.32 mm**

Spider Web

Parameter	Value
N_r	**15**
N_s	**18**
t_r	**2 mm**
t_f	**10 mm**
σ_j	**-0.0035939**
A	**1.6913**
f	**90.018**
ϕ	**58.93**
p	**0.1149**

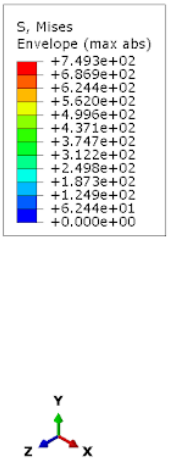
Area fraction: **56.58%** · Shell thickness: **8.27 mm**

1.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	749.34	542.58
Total ESEDEN (MPa)	9710.77	6263.92
Max ESEDEN (MPa)	0.5390	0.2459
Max displacement (mm)	2.8439	2.4717

Table 1. *Performance metrics — case 1 (SS · Uniform · von Mises).*

1.3. Field Contours



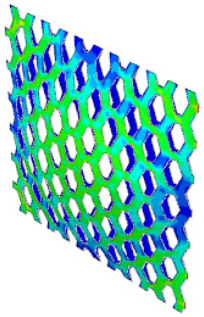
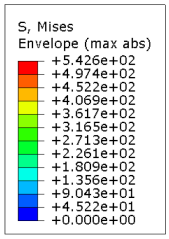


Figure 1. *Honeycomb — optimal field.*



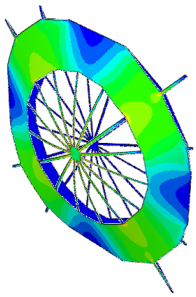




Figure 2. *Spider web — optimal field.*

1.4. Optimization Convergence with Uncertainty Quantification

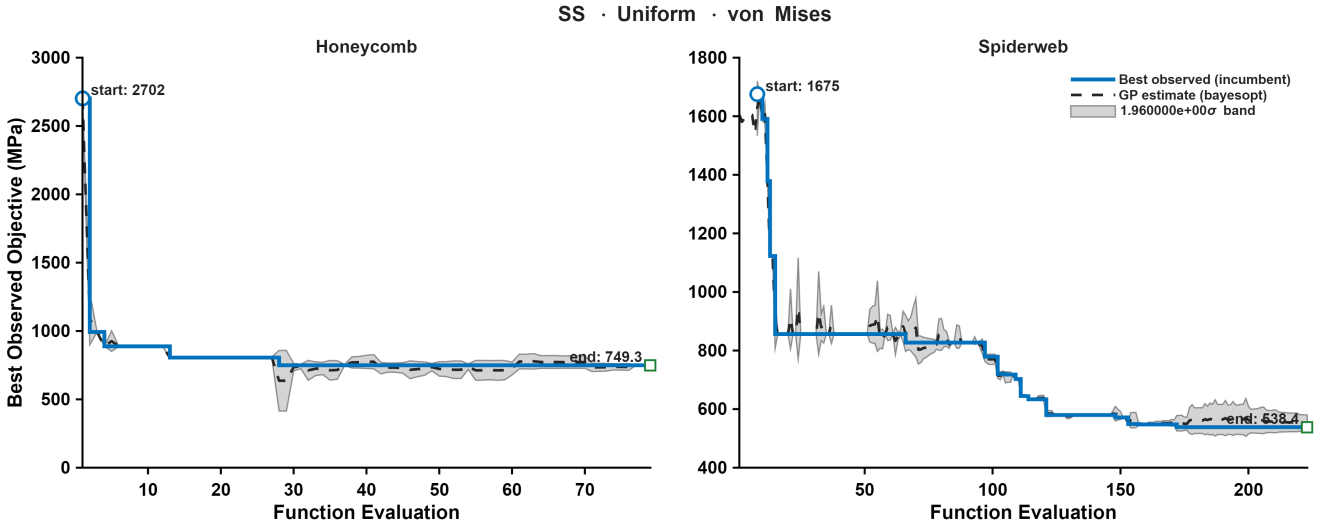


Figure 3. *Bayesian optimization convergence history for case 1 (SS · Uniform · von Mises). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

2. SS · Uniform · Total ESEDEN

BC: Simply supported · Load: Uniform pressure · Objective: Minimize total ESEDEN

Spider Web wins +40%

2.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.0206**
β	**0.43161**
t	**9 mm**
h	**39.009 mm**

Area fraction: **59.03%** · Shell thickness: **8.39 mm**

Spider Web

Parameter	Value
N_r	**16**
N_s	**9**
t_r	**2 mm**
t_f	**11 mm**
σ_j	**-0.23455**
A	**6.5148**
f	**-31.241**
ϕ	**-84.589**
p	**0.10332**

Area fraction: **59.43%** · Shell thickness: **8.41 mm**

2.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	740.86	731.07
Total ESEDEN (MPa)	8676.57	5224.40
Max ESEDEN (MPa)	0.7952	0.4873
Max displacement (mm)	2.5507	2.2074

Table 2. *Performance metrics — case 2 (SS · Uniform · Total ESEDEN).*

2.3. Field Contours

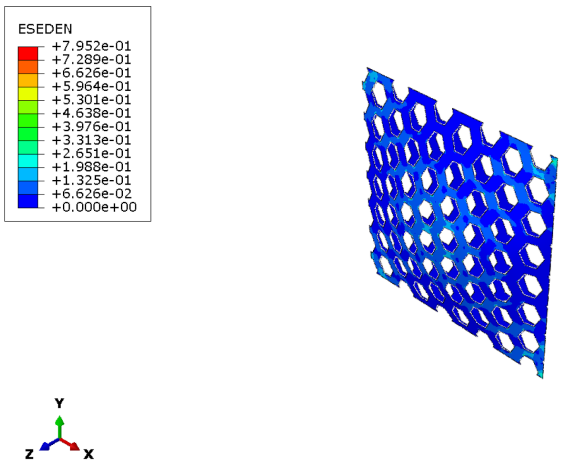


Figure 4. *Honeycomb — optimal field.*

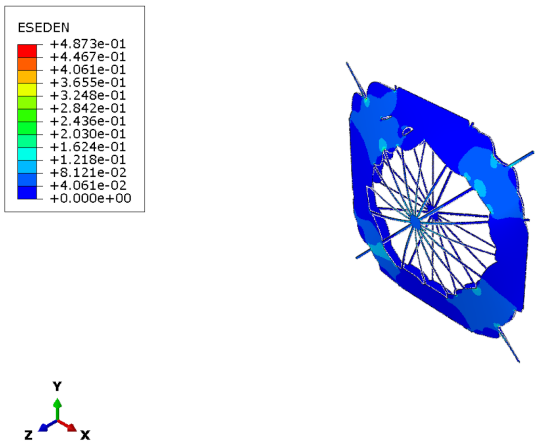


Figure 5. *Spider web — optimal field.*

2.4. Optimization Convergence with Uncertainty Quantification

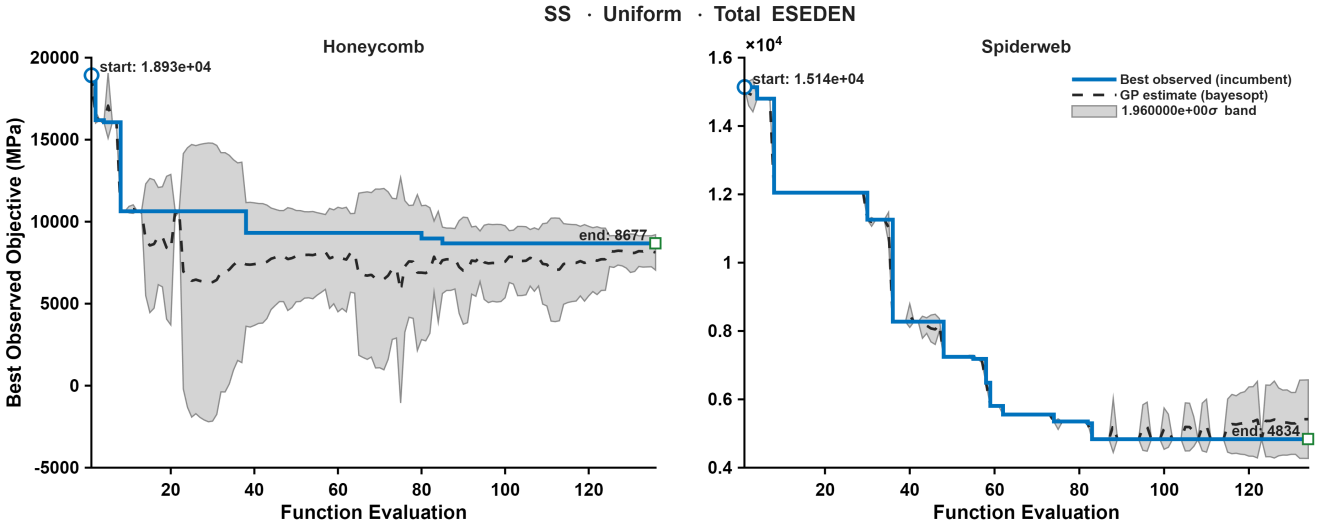


Figure 6. *Bayesian optimization convergence history for case 2 (SS · Uniform · Total ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

3. SS · Uniform · Max ESEDEN

BC: Simply supported · Load: Uniform pressure · Objective: Minimize maximum ESEDEN

Spider Web wins +35%

3.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.1483**
β	**0.25491**
t	**7 mm**
h	**30.162 mm**

Area fraction: **56.26%** · Shell thickness: **8.26 mm**

Spider Web

Parameter	Value
N_r	**16**
N_s	**13**
t_r	**3 mm**
t_f	**6 mm**
σ_j	**0.043577**
A	**5.8797**
f	**-92.733**
ϕ	**-68.386**
p	**0.13227**

Area fraction: **56.14%** · Shell thickness: **8.25 mm**

3.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	763.52	700.79
Total ESEDEN (MPa)	9653.43	7205.85
Max ESEDEN (MPa)	0.5780	0.3740
Max displacement (mm)	2.9314	2.6160

Table 3. *Performance metrics — case 3 (SS · Uniform · Max ESEDEN).*

3.3. Field Contours

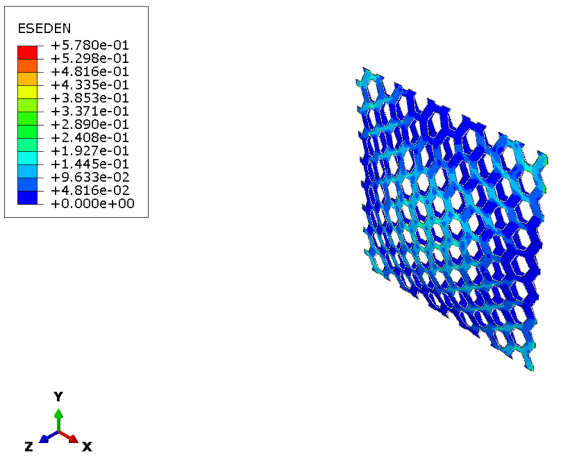


Figure 7. *Honeycomb — optimal field.*

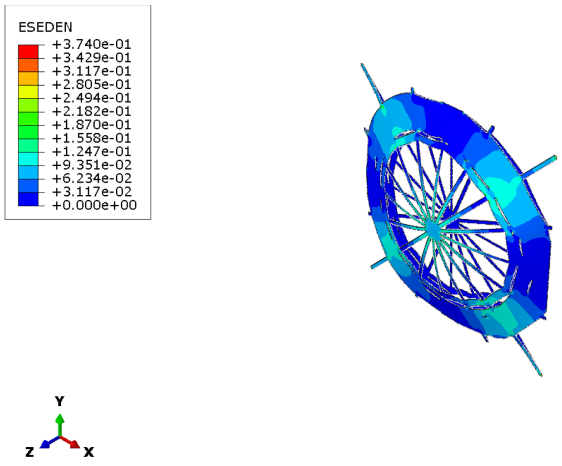


Figure 8. *Spider web — optimal field.*

3.4. Optimization Convergence with Uncertainty Quantification

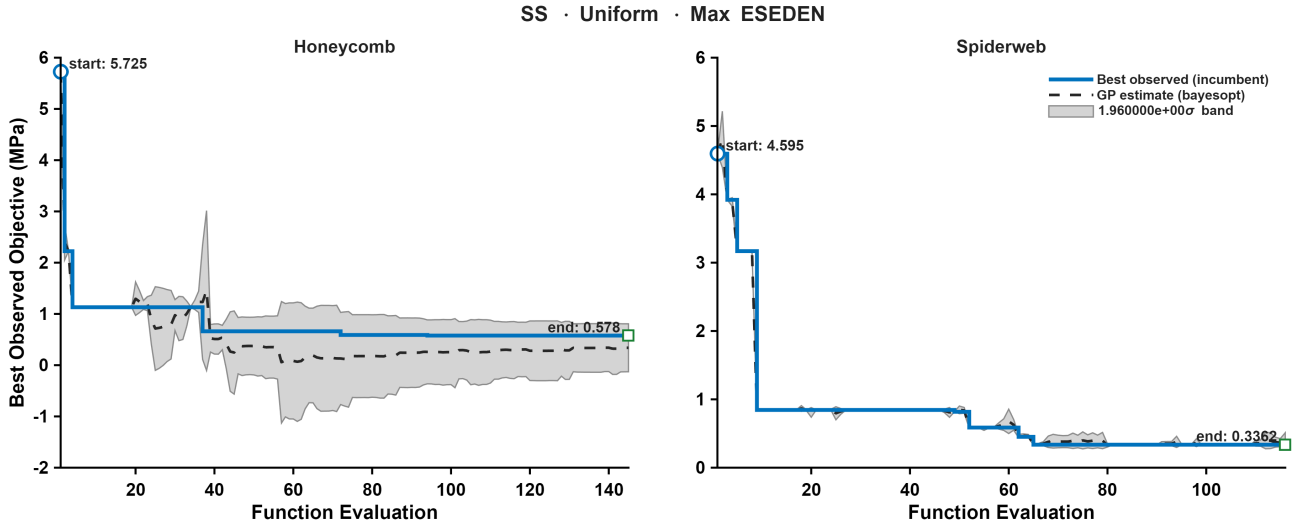


Figure 9. *Bayesian optimization convergence history for case 3 (SS · Uniform · Max ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

4. SS · Non-Uniform · von Mises

BC: Simply supported · Load: Non-uniform pressure · Objective: Minimize maximum von Mises stress

Honeycomb wins +4%

4.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.2688**
β	**0.69557**
t	**8 mm**
h	**30.217 mm**

Area fraction: **59.87%** · Shell thickness: **8.43 mm**

Spider Web

Parameter	Value
N_r	**12**
N_s	**3**
t_r	**10 mm**
t_f	**10 mm**
σ_j	**0.025575**
A	**3.5842**
f	**81.4**
ϕ	**52.079**
p	**0.1051**

Area fraction: **55.27%** · Shell thickness: **8.21 mm**

4.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	350.62	364.10
Total ESEDEN (MPa)	1356.93	1496.47
Max ESEDEN (MPa)	0.1140	0.1434
Max displacement (mm)	0.9913	1.0953

Table 4. *Performance metrics — case 4 (SS · Non-Uniform · von Mises).*

4.3. Field Contours

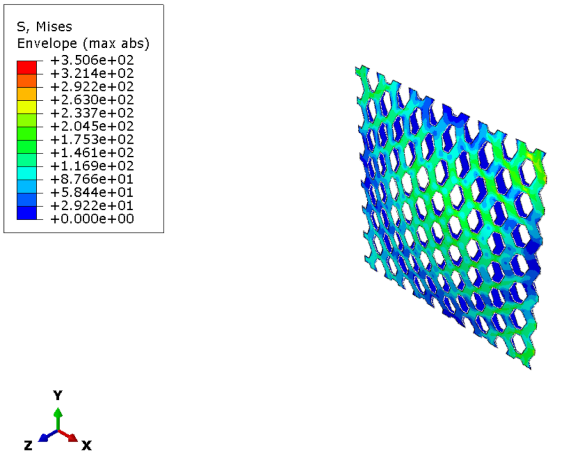


Figure 10. *Honeycomb — optimal field.*

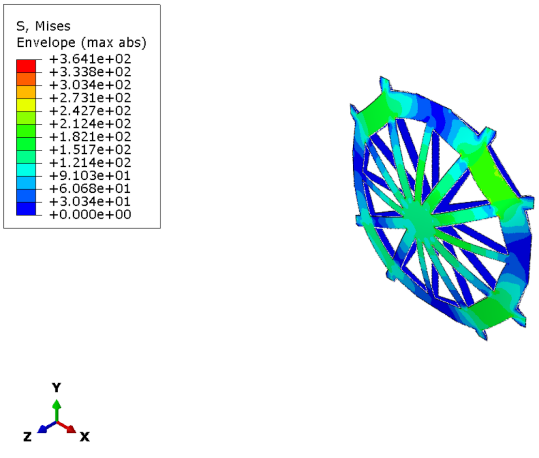


Figure 11. *Spider web — optimal field.*

4.4. Optimization Convergence with Uncertainty Quantification

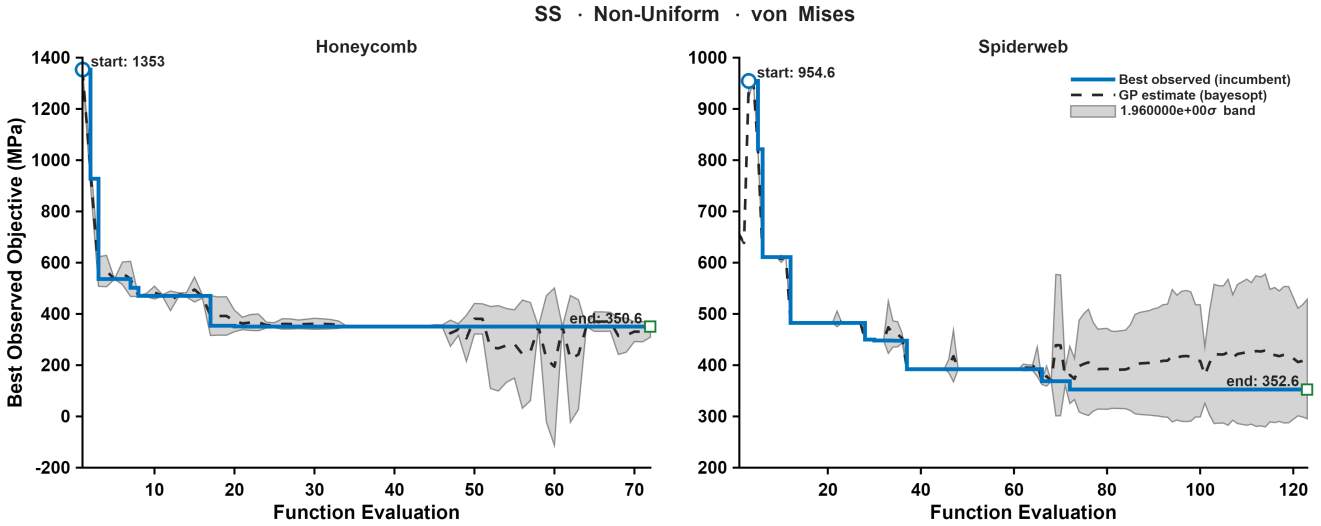


Figure 12. *Bayesian optimization convergence history for case 4 (SS · Non-Uniform · von Mises). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

5. SS · Non-Uniform · Total ESEDEN

BC: Simply supported · Load: Non-uniform pressure · Objective: Minimize total ESEDEN

Spider Web wins +22%

5.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.1093**
β	**0.3425**
t	**9 mm**
h	**37.672 mm**

Area fraction: **58.16%** · Shell thickness: **8.35 mm**

Spider Web

Parameter	Value
N_r	**16**
N_s	**5**
t_r	**2 mm**
t_f	**11 mm**
σ_j	**0.11192**
A	**8.6247**
f	**45.702**
ϕ	**90.51**
p	**0.12898**

Area fraction: **49.17%** · Shell thickness: **7.89 mm**

5.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	378.48	434.91
Total ESEDEN (MPa)	1324.85	1033.76
Max ESEDEN (MPa)	0.1145	0.1484
Max displacement (mm)	0.9988	1.1244

Table 5. *Performance metrics — case 5 (SS · Non-Uniform · Total ESEDEN).*

5.3. Field Contours

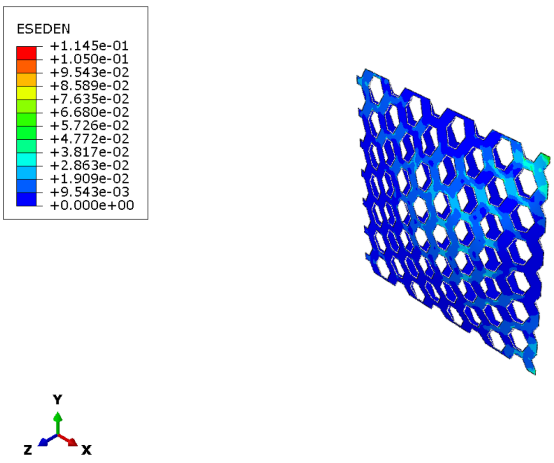


Figure 13. *Honeycomb — optimal field.*

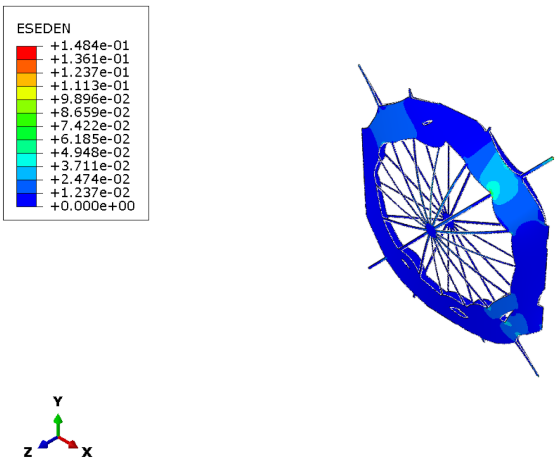


Figure 14. *Spider web — optimal field.*

5.4. Optimization Convergence with Uncertainty Quantification

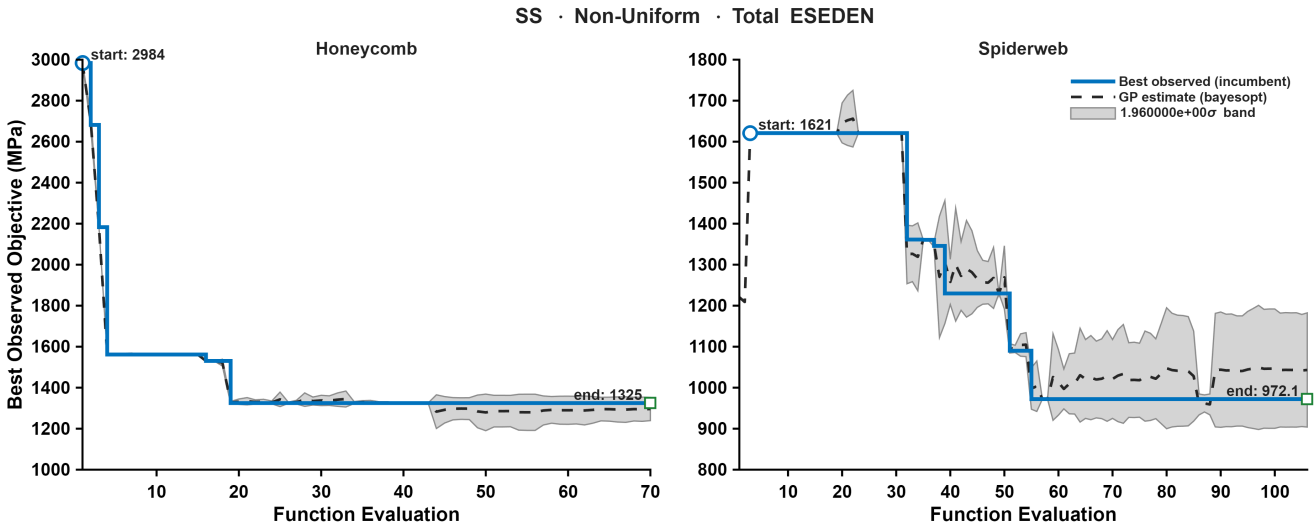


Figure 15. *Bayesian optimization convergence history for case 5 (SS · Non-Uniform · Total ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

6. SS · Non-Uniform · Max ESEDEN

BC: Simply supported · Load: Non-uniform pressure · Objective: Minimize maximum ESEDEN

Honeycomb wins +37%

6.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.0139**
β	**0.22209**
t	**7 mm**
h	**30.374 mm**

Area fraction: **58.91%** · Shell thickness: **8.38 mm**

Spider Web

Parameter	Value
N_r	**12**
N_s	**23**
t_r	**12 mm**
t_f	**3 mm**
σ_j	**0.022263**
A	**4.0188**
f	**-17.492**
ϕ	**96.485**
p	**2.8646**

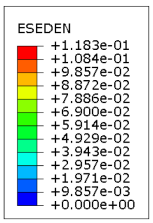
Area fraction: **54.89%** · Shell thickness: **8.19 mm**

6.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	403.63	433.48
Total ESEDEN (MPa)	1367.19	1908.52
Max ESEDEN (MPa)	0.1183	0.1868
Max displacement (mm)	1.0272	1.0971

Table 6. *Performance metrics — case 6 (SS · Non-Uniform · Max ESEDEN).*

6.3. Field Contours



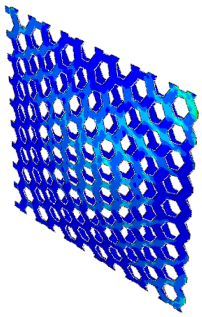
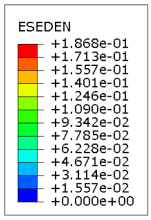




Figure 16. *Honeycomb — optimal field.*



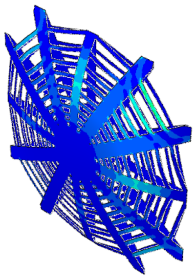




Figure 17. *Spider web — optimal field.*

6.4. Optimization Convergence with Uncertainty Quantification

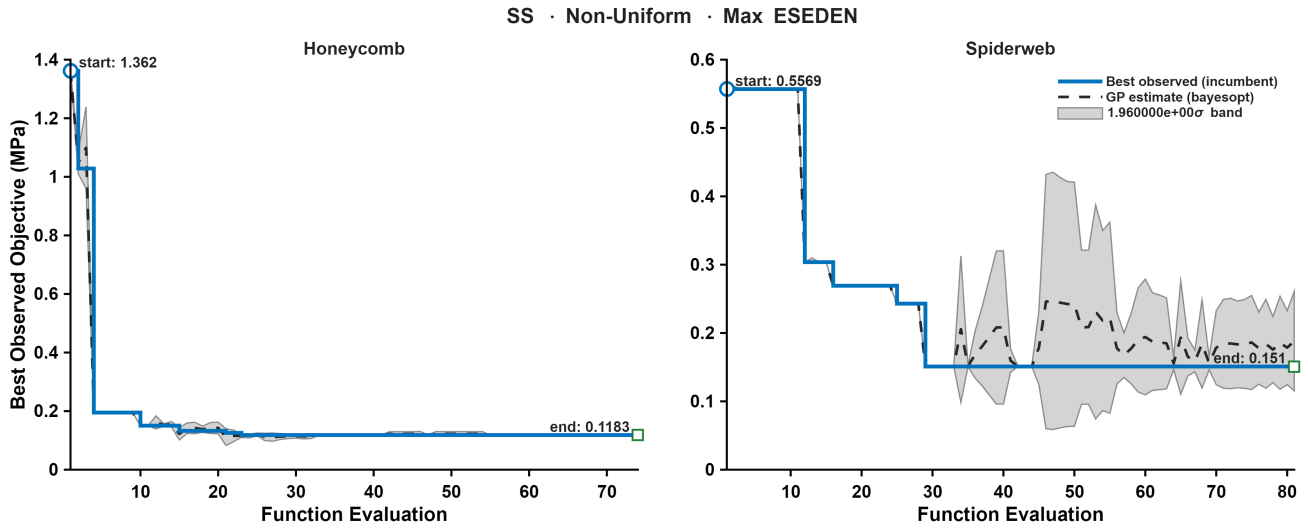


Figure 18. *Bayesian optimization convergence history for case 6 (SS · Non-Uniform · Max ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

7. CC · Uniform · von Mises

BC: Clamped · Load: Uniform pressure · Objective: Minimize maximum von Mises stress

Equivalent Performance ~0.2%

7.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.1045**
β	**0.11402**
t	**9 mm**
h	**35.294 mm**

Area fraction: **59.89%** · Shell thickness: **8.43 mm**

Spider Web

Parameter	Value
N_r	**16**
N_s	**3**
t_r	**11 mm**
t_f	**12 mm**
σ_j	**0.030589**
A	**0.20727**
f	**71.758**
ϕ	**10.992**
p	**2.4755**

Area fraction: **59.85%** · Shell thickness: **8.43 mm**

7.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	639.72	641.23
Total ESEDEN (MPa)	2062.67	2437.37
Max ESEDEN (MPa)	0.3890	0.4123
Max displacement (mm)	0.7876	0.7189

Table 7. *Performance metrics — case 7 (CC · Uniform · von Mises).*

7.3. Field Contours

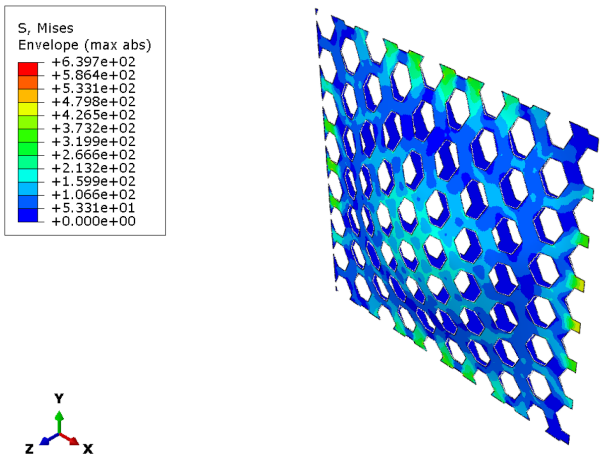


Figure 19. *Honeycomb — optimal field.*

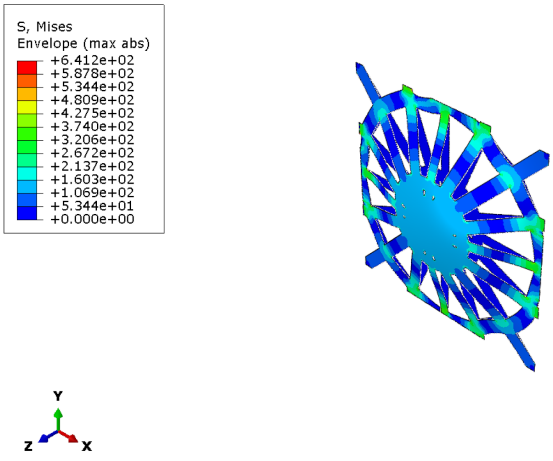


Figure 20. *Spider web — optimal field.*

7.4. Optimization Convergence with Uncertainty Quantification

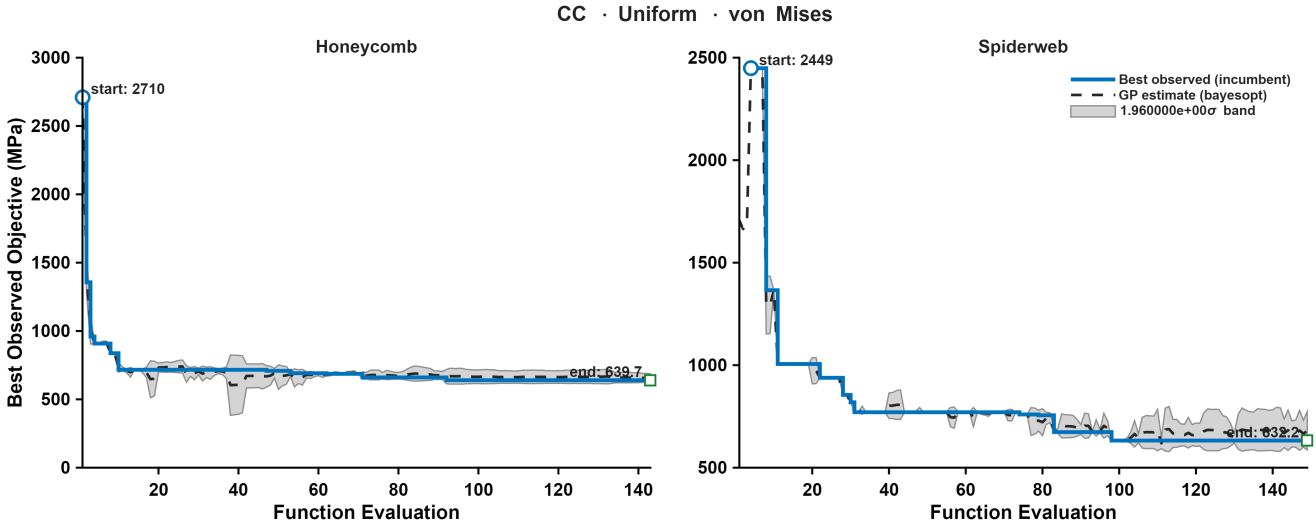


Figure 21. *Bayesian optimization convergence history for case 7 (CC · Uniform · von Mises). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

8. CC · Uniform · Total ESEDEN

BC: Clamped · Load: Uniform pressure · Objective: Minimize total ESEDEN

Spider Web wins +73%

8.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.0338**
β	**0.15278**
t	**7 mm**
h	**31.742 mm**

Area fraction: **57.05%** · Shell thickness: **8.29 mm**

Spider Web

Parameter	Value
N_r	**12**
N_s	**23**
t_r	**2 mm**
t_f	**7 mm**
σ_j	**-0.16330**
A	**1.0835**
f	**-3.2448**
ϕ	**-87.266**
p	**0.10426**

Area fraction: **57.49%** · Shell thickness: **8.32 mm**

8.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	768.85	496.84
Total ESEDEN (MPa)	2089.54	570.79
Max ESEDEN (MPa)	0.5564	0.2319
Max displacement (mm)	0.8699	0.4626

Table 8. *Performance metrics — case 8 (CC · Uniform · Total ESEDEN).*

8.3. Field Contours

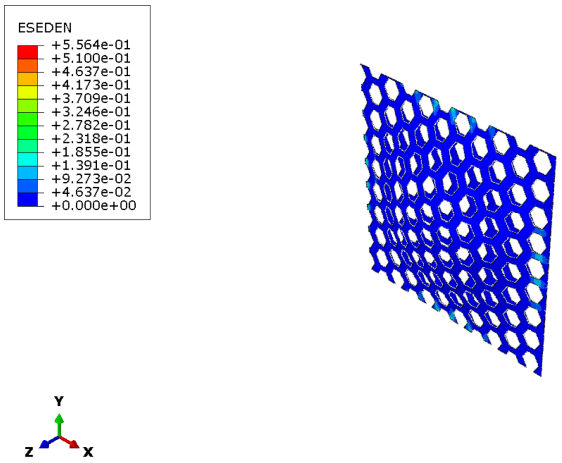


Figure 22. *Honeycomb — optimal field.*

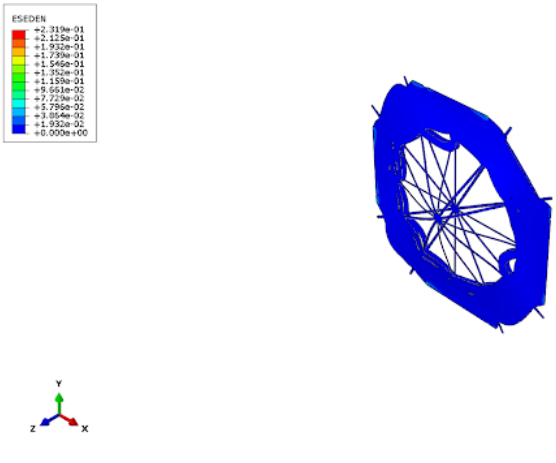


Figure 23. *Spider web — optimal field.*

8.4. Optimization Convergence with Uncertainty Quantification

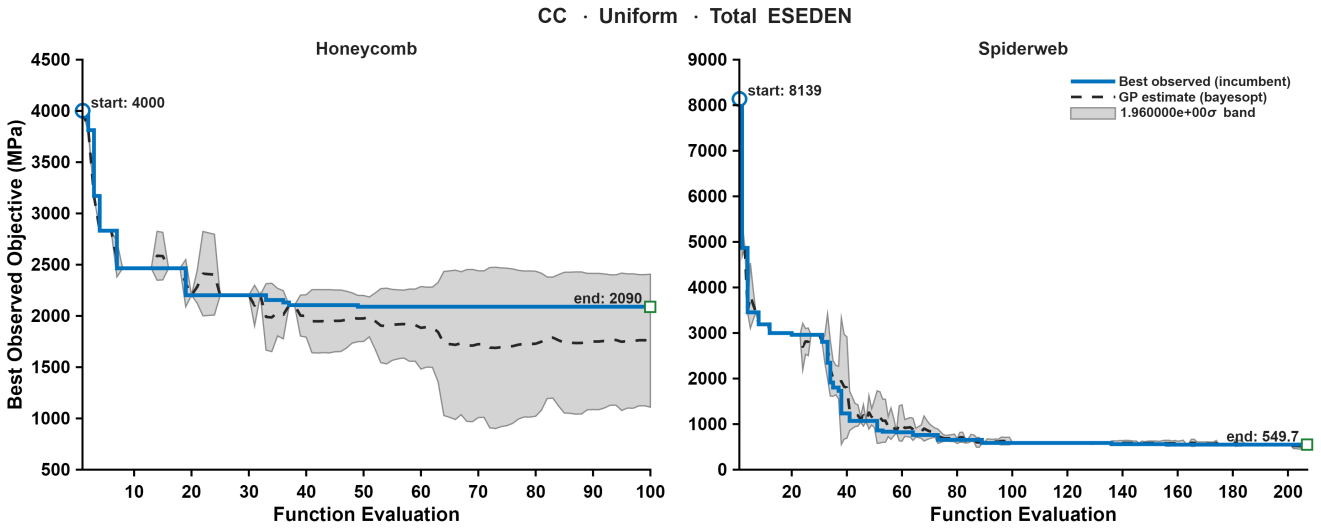


Figure 24. *Bayesian optimization convergence history for case 8 (CC · Uniform · Total ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

9. CC · Uniform · Max ESEDEN

BC: Clamped · Load: Uniform pressure · Objective: Minimize maximum ESEDEN

Honeycomb wins +37%

9.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**0.77307**
β	**0.10294**
t	**8 mm**
h	**78.251 mm**

Area fraction: **57.63%** · Shell thickness: **8.32 mm**

Spider Web

Parameter	Value
N_r	**23**
N_s	**3**
t_r	**10 mm**
t_f	**3 mm**
σ_j	**-0.13900**
A	**6.6163**
f	**17.199**
ϕ	**92.25**
p	**2.7102**

Area fraction: **59.75%** · Shell thickness: **8.42 mm**

9.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	644.38	770.46
Total ESEDEN (MPa)	2244.17	3183.36
Max ESEDEN (MPa)	0.3858	0.6112
Max displacement (mm)	0.9066	0.8306

Table 9. *Performance metrics — case 9 (CC · Uniform · Max ESEDEN).*

9.3. Field Contours

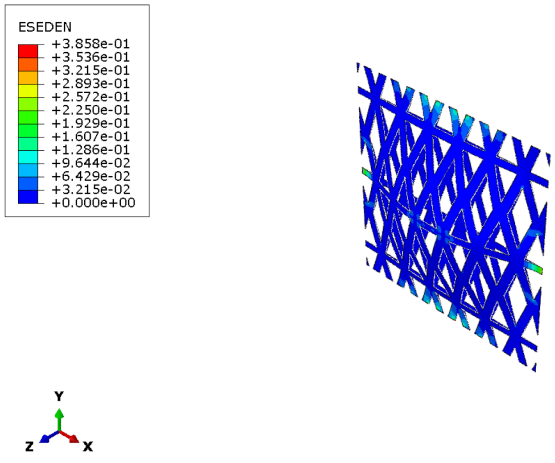


Figure 25. *Honeycomb — optimal field.*

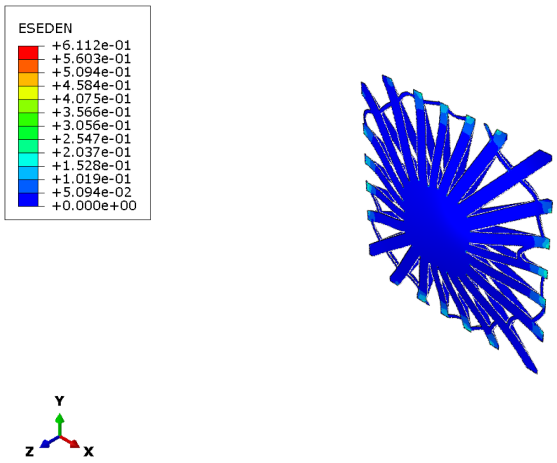


Figure 26. *Spider web — optimal field.*

9.4. Optimization Convergence with Uncertainty Quantification

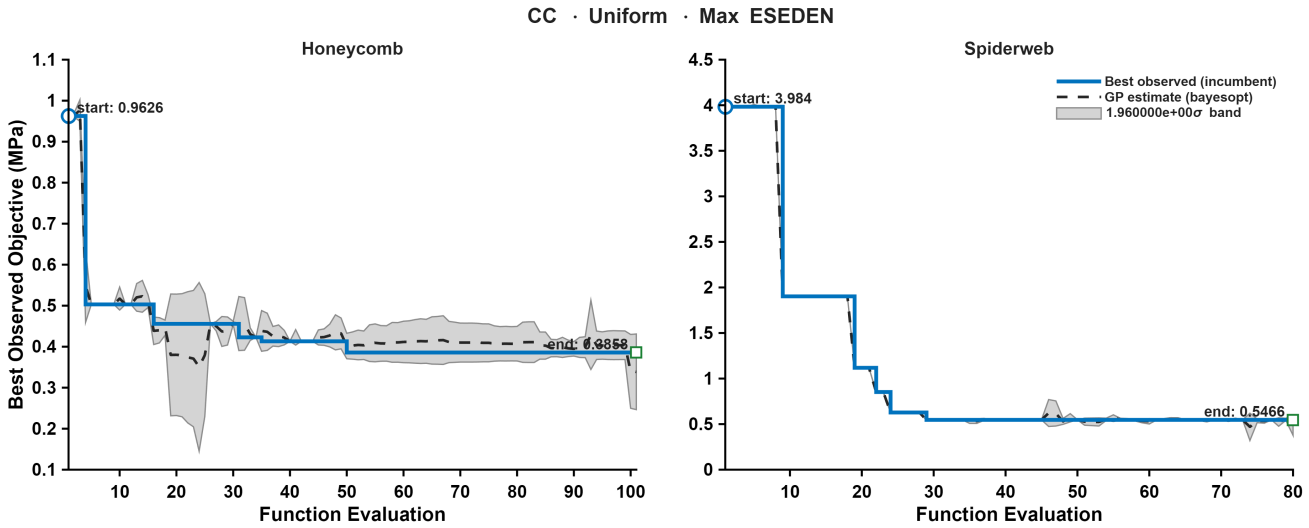


Figure 27. *Bayesian optimization convergence history for case 9 (CC · Uniform · Max ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

10. CC · Non-Uniform · von Mises

BC: Clamped · Load: Non-uniform pressure · Objective: Minimize maximum von Mises stress

Honeycomb wins +16%

10.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.5759**
β	**0.73028**
t	**9 mm**
h	**31.281 mm**

Area fraction: **59.77%** · Shell thickness: **8.42 mm**

Spider Web

Parameter	Value
N_r	**18**
N_s	**2**
t_r	**6 mm**
t_f	**12 mm**
σ_j	**0.17202**
A	**7.7598**
f	**64.030**
ϕ	**6.784**
p	**2.9218**

Area fraction: **55.27%** · Shell thickness: **8.21 mm**

10.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	272.80	325.84
Total ESEDEN (MPa)	288.68	212.55
Max ESEDEN (MPa)	0.0801	0.1011
Max displacement (mm)	0.2966	0.2662

Table 10. *Performance metrics — case 10 (CC · Non-Uniform · von Mises).*

10.3. Field Contours

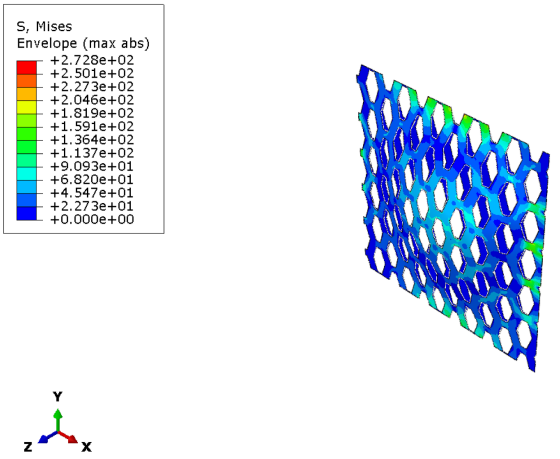


Figure 28. *Honeycomb — optimal field.*

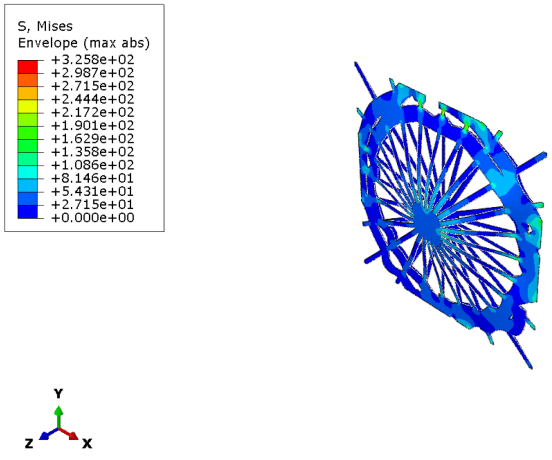


Figure 29. *Spider web — optimal field.*

10.4. Optimization Convergence with Uncertainty Quantification

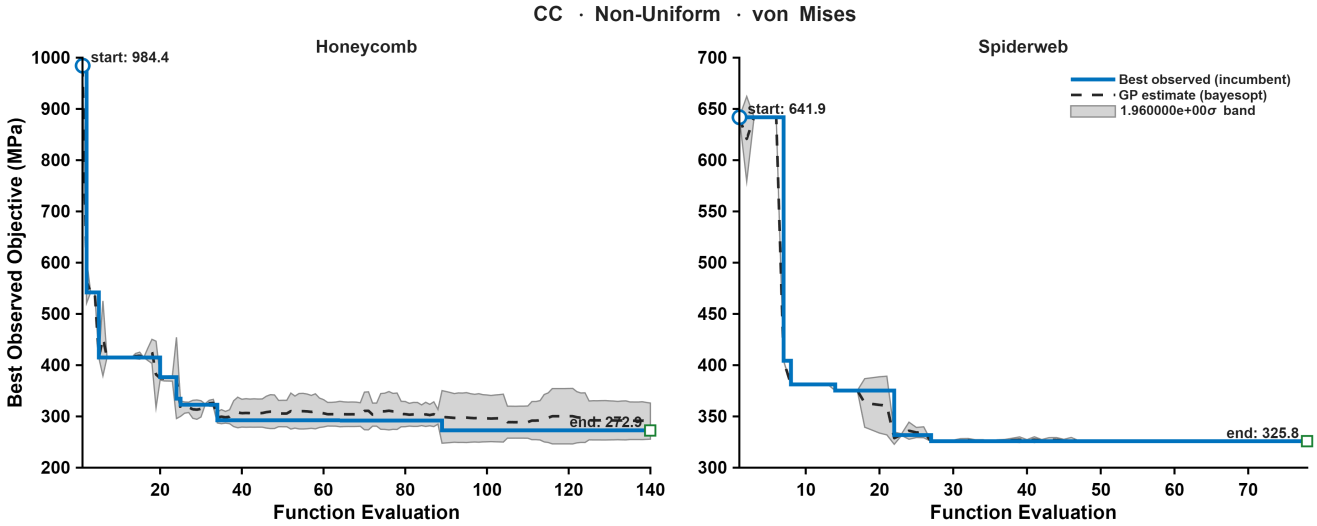


Figure 30. *Bayesian optimization convergence history for case 10 (CC · Non-Uniform · von Mises). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

11. CC · Non-Uniform · Total ESEDEN

BC: Clamped · Load: Non-uniform pressure · Objective: Minimize total ESEDEN

Spider Web wins +57%

11.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.0059**
β	**0.15311**
t	**9 mm**
h	**38.647 mm**

Area fraction: **59.35%** · Shell thickness: **8.40 mm**

Spider Web

Parameter	Value
N_r	**12**
N_s	**9**
t_r	**2 mm**
t_f	**10 mm**
σ_j	**0.28544**
A	**1.1621**
f	**-18.272**
ϕ	**0.6142**
p	**0.11172**

Area fraction: **60.18%** · Shell thickness: **8.44 mm**

11.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	333.09	337.42
Total ESEDEN (MPa)	269.87	115.30
Max ESEDEN (MPa)	0.1044	0.1070
Max displacement (mm)	0.2826	0.1805

Table 11. *Performance metrics — case 11 (CC · Non-Uniform · Total ESEDEN).*

11.3. Field Contours

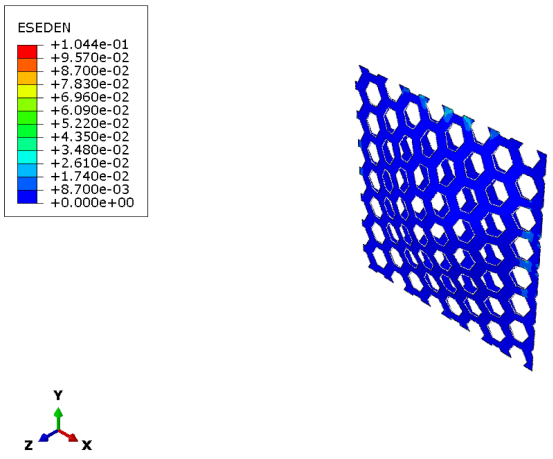


Figure 31. *Honeycomb — optimal field.*

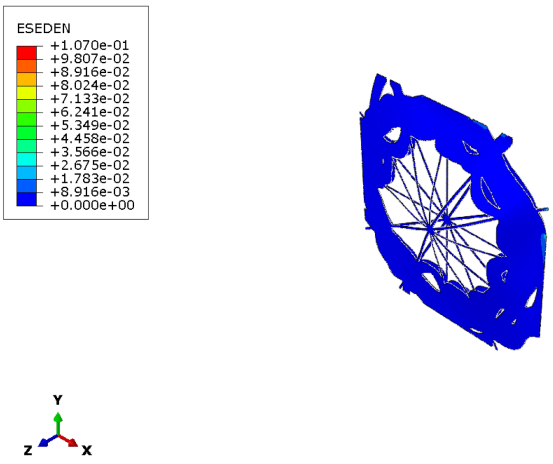


Figure 32. *Spider web — optimal field.*

11.4. Optimization Convergence with Uncertainty Quantification

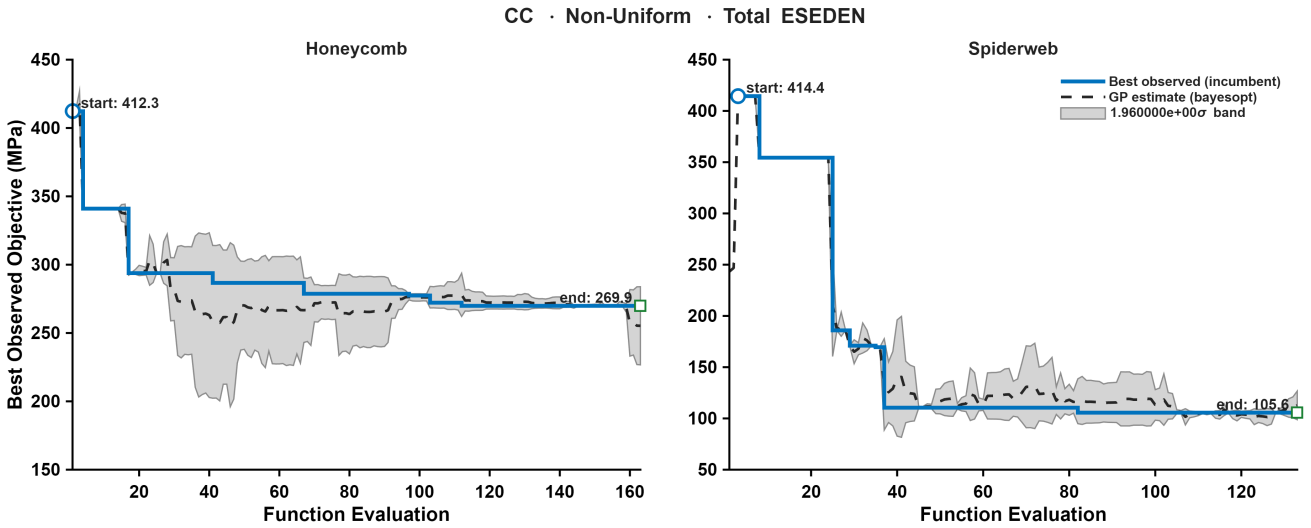


Figure 33. *Bayesian optimization convergence history for case 11 (CC · Non-Uniform · Total ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*

12. CC · Non-Uniform · Max ESEDEN

BC: Clamped · Load: Non-uniform pressure · Objective: Minimize maximum ESEDEN

Honeycomb wins +38%

12.1. Optimal Design Parameters

Honeycomb

Parameter	Value
α	**1.1682**
β	**0.13938**
t	**9 mm**
h	**35.094 mm**

Area fraction: **59.43%** · Shell thickness: **8.41 mm**

Spider Web

Parameter	Value
N_r	**22**
N_s	**2**
t_r	**10 mm**
t_f	**4 mm**
σ_j	**0.15416**
A	**7.2527**
f	**-55.571**
ϕ	**56.716**
p	**0.90885**

Area fraction: **59.91%** · Shell thickness: **8.43 mm**

12.2. Quantitative Results

Metric	Honeycomb	Spider Web
Max von Mises stress (MPa)	290.81	362.93
Total ESEDEN (MPa)	298.62	412.82
Max ESEDEN (MPa)	0.0765	0.1241
Max displacement (mm)	0.2933	0.2915

Table 12. *Performance metrics — case 12 (CC · Non-Uniform · Max ESEDEN).*

12.3. Field Contours

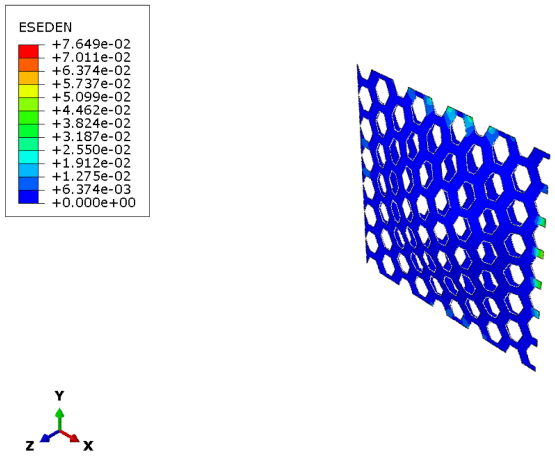


Figure 34. *Honeycomb — optimal field.*

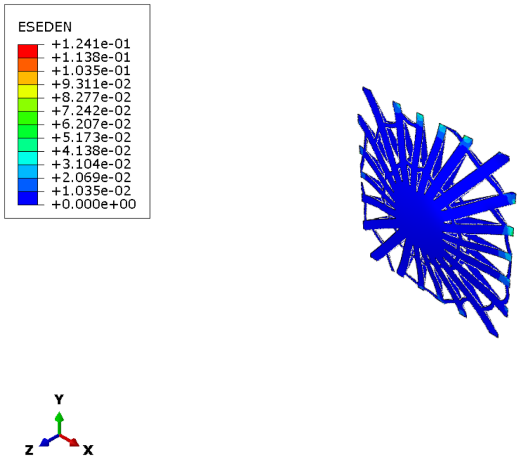


Figure 35. *Spider web — optimal field.*

12.4. Optimization Convergence with Uncertainty Quantification

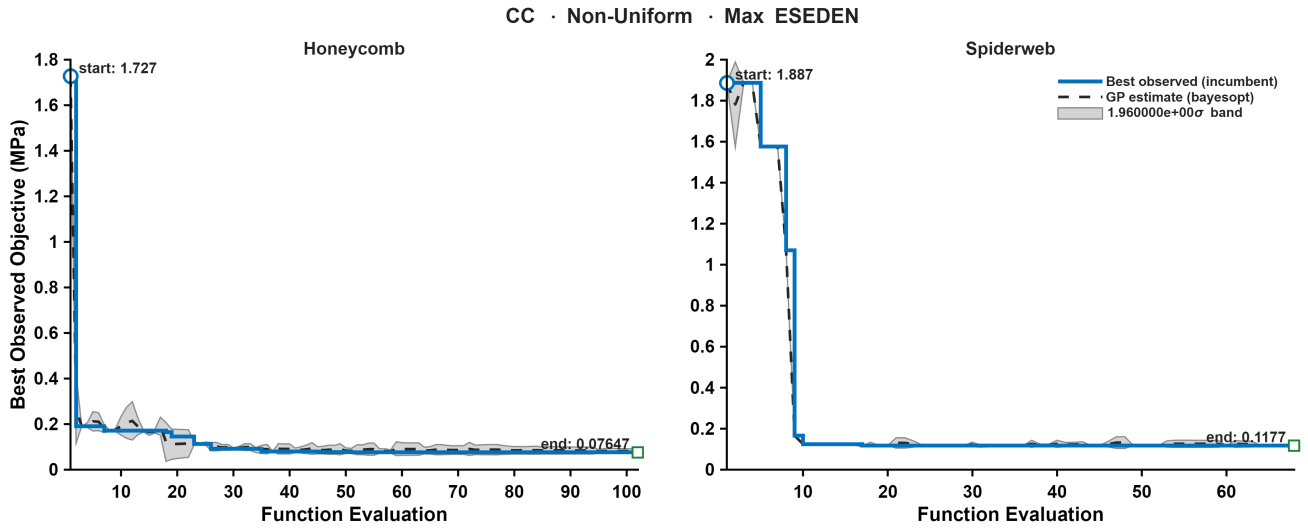


Figure 36. *Bayesian optimization convergence history for case 12 (CC · Non-Uniform · Max ESEDEN). Left panel: honeycomb; right panel: spider web. Solid line — best observed incumbent objective; dashed line — Gaussian-process posterior mean; shaded band — 1.96σ credible interval.*